

# STEPHEN MASON

Nairobi, Kenya | [masonstephen606@gmail.com](mailto:masonstephen606@gmail.com) | +254 712 382 482

[Portfolio: stephen-mason.tech](https://stephen-mason.tech) | [DataViz: datavizual.app](https://datavizual.app) | [LinkedIn: linkedin.com/in/stephen-mason-ab494a2b3](https://linkedin.com/in/stephen-mason-ab494a2b3) | [GitHub: github.com/masonstephen](https://github.com/masonstephen)

## PROFESSIONAL SUMMARY

Final-year Bachelor of Business Information Technology student at Strathmore University with hands-on experience in software engineering, data-driven applications, system testing, and technical support. Built and independently deployed DataViz, a live data visualization platform, and designed a curriculum-aware retrieval-augmented generation (RAG) research proposal (LibAI) for trustworthy educational AI. Seeking graduate research and professional opportunities in Artificial Intelligence, Data Science, or Computer Science, with a focus on natural language processing, information retrieval, and evidence-grounded AI systems.

## RESEARCH INTERESTS

Artificial Intelligence and Machine Learning; Natural Language Processing and Large Language Models; Retrieval-Augmented Generation (RAG) and Information Retrieval; Trustworthy AI, Evidence Grounding, and Evaluation; Data Science and Data Mining.

## TECHNICAL SKILLS

Programming Languages: Python, PHP, JavaScript, TypeScript

Frontend Development: React, Vue.js, Next.js

Backend Development: FastAPI, Django, Laravel, Supabase

Databases: PostgreSQL, MySQL, Supabase

Data & AI: Pandas, NumPy, data visualization and analytics, retrieval-augmented generation (RAG) concepts, prompt engineering

Tools: Git, GitHub, REST APIs, Docker (basic), Microsoft Word, PowerPoint, Excel

Additional Learning: Solidity and smart-contract concepts

## EDUCATION

**Bachelor of Business Information Technology** **Apr 2024 – Jun 2027 (expected)**  
*Strathmore University* *Nairobi, Kenya*

**Diploma in Business Information Technology** **Completed Oct 2025**  
*Strathmore University — embedded diploma, completed en route to bachelor's degree* *Nairobi, Kenya*

## RESEARCH AND TECHNICAL PROJECTS

### LibAI — Curriculum-Aware RAG for Educational Question Answering

*Independent Research & Development*

*In Progress*

- Designed a curriculum-aware retrieval-augmented generation (RAG) architecture for Liberian secondary education, comparing BM25, dense, and hybrid retrieval strategies against curriculum metadata (grade, subject, topic, learning objective).
- Authored a three-chapter research proposal (Design Science Research methodology) defining research questions, six controlled experimental configurations, and an evaluation framework covering retrieval quality (Precision@K, Recall@K, MRR), faithfulness, citation correctness, and answerability.
- Currently building the curriculum knowledge base and vector index ahead of implementation.
- Technologies: Python, React, retrieval-augmented generation, large language models.

### DataViz — AI-Assisted Data Visualization Platform

**2026, Live**

- Independently built and deployed a full-stack data visualization and dashboard platform, live at [datavizual.app](https://datavizual.app) with 11+ active users.
- Delivered dataset upload and exploration, 30+ interactive chart types, AI-assisted chart recommendations, and export to PDF, PNG, and Excel.
- Technologies: FastAPI, Python, PostgreSQL, Pandas, NumPy, React, TypeScript, Vite, Tailwind CSS, ECharts.

## **Gitaru Market Linkage Initiative — Academic Information Systems Project**

*Collaborative Group of Four*

*Ongoing*

- Investigating market-access challenges faced by smallholder farmers at Gitaru Market, including intermediaries, buyer demand, market information, transportation, and post-harvest losses, as part of a four-person team.
- Applied Design Thinking and Participatory Action Research to assess stakeholder needs and evaluate market-linkage interventions; selected farmer cooperatives as the proposed intervention.
- Project presentation scheduled for 25 November 2026.

### **LibTicket — Digital Event Ticketing Platform**

**In Development**

- Developing a platform for event creation, digital ticket issuance, attendee registration, and QR-code-based ticket validation.
- Technologies: Vue.js, Supabase, JavaScript, REST APIs.

### **LibLand — Blockchain Land Registry Concept**

**Concept Stage**

- Scoping a blockchain-based approach to land ownership records to improve transparency and reduce fraud and disputes; evaluating smart-contract architectures for land registration and ownership transfer (Solidity, learning in progress).

## **PROFESSIONAL EXPERIENCE**

---

### **Software Intern**

**Feb 2026 – May 2026**

*iBiz Africa (iLab Africa), Strathmore University*

*Nairobi, Kenya*

- Contributed to the design, development, and testing of digital systems, including UI/UX improvements and system analysis for the Virtual Traffic Court (VTC).
- Carried out quality assurance testing to identify and resolve bugs ahead of release, supporting system reliability.
- Collaborated with cross-functional team members to translate business requirements into technical solutions.

### **Computer Lab Assistant**

**May 2024 – Feb 2026**

*Strathmore University*

*Nairobi, Kenya*

- Resolved technical issues and system-access problems for students and staff, supporting daily computing laboratory operations.
- Troubleshoot network issues, installed and configured software, and maintained lab computers.
- Maintained reliable access to computing resources for lecturers and students during practical class sessions.

### **Assistant Computer Teacher**

**Apr 2021 – Sep 2023**

*Dr. Abraham S. Borbor Memorial School of Excellence*

*Paynesville, Liberia*

- Delivered hands-on technical guidance to students to strengthen digital literacy, including an introductory lesson on AI and machine learning that shaped my current research interests.
- Prepared lesson materials and troubleshoot classroom hardware and software issues.

## **CERTIFICATIONS**

---

- Cisco Networking Academy — CCNA: Switching, Routing, and Wireless Essentials
- Cisco Networking Academy — Introduction to Data Science
- HP LIFE — Data Science & Analytics, Level 1
- HP LIFE — Presenting Data, Level 1
- Teacher Impact and Empowerment Training — Certificate of Completion, Level 1
- EF SET English Certificate — 63/100, C1 Advanced (Reading C1, Listening C2, Writing B2, Speaking B2)

## **REFERENCES**

---

Available upon request.